Numpy cheatsheet

NumPy is the fundamental package for scientific computing with Python. This cheat sheet is a quick reference for NumPy beginners.

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Getting Started

Introduction

You'll also need to import numpy to get started:

```
import numpy as np
```

Importing/exporting

`np.loadtxt('file.txt')`	From a text file
`np.genfromtxt('file.csv',delimiter=',')`	From a CSV file
`np.savetxt('file.txt',arr,delimiter=' ')`	Writes to a text file
`np.savetxt('file.csv',arr,delimiter=',')`	Writes to a CSV file

Creating Arrays

`np.array([1,2,3])`	One dimensional array
`np.array([(1,2,3),(4,5,6)])`	Two dimensional array
`np.zeros(3)`	1D array of length 3 all values 0
`np.ones((3,4))`	3x4 array with all values 1
`np.eye(5)`	5x5 array of 0 with 1 on diagonal (Identity matrix)
`np.linspace(0,100,6)`	Array of 6 evenly divided values from 0 to 100
`np.arange(0,10,3)`	Array of values from 0 to less than 10 with step 3 (eg [0,3,6,9])
`np.full((2,3),8)`	2x3 array with all values 8
`np.random.rand(4,5)`	4x5 array of random floats between 0–1
`np.random.rand(6,7)*100`	6x7 array of random floats between 0–100
`np.random.randint(5,size=(2,3))`	2x3 array with random ints between 0–4

Inspecting Properties

`arr.size`	Returns number of elements in arr
`arr.shape`	Returns dimensions of arr (rows,columns)
`arr.dtype`	Returns type of elements in arr
`arr.astype(dtype)`	Convert arr elements to type dtype
`arr.tolist()`	Convert arr to a Python list
`np.info(np.eye)`	View documentation for np.eye

Copying/sorting/reshaping

`np.copy(arr)`	Copies arr to new memory
`arr.view(dtype)`	Creates view of arr elements with type dtype
`arr.sort()`	Sorts arr
`arr.sort(axis=0)`	Sorts specific axis of arr
`two_d_arr.flatten()`	Flattens 2D array two_d_arr to 1D
`arr.T`	Transposes arr (rows become columns and vice versa)
`arr.reshape(3,4)`	Reshapes arr to 3 rows, 4 columns without changing data
`arr.resize((5,6))`	Changes arr shape to 5x6 and fills new values with 0

Adding/removing Elements

`np.append(arr,values)`	Appends values to end of arr
`np.insert(arr,2,values)`	Inserts values into arr before index 2
`np.delete(arr,3,axis=0)`	Deletes row on index 3 of arr
`np.delete(arr,4,axis=1)`	Deletes column on index 4 of arr

Combining/splitting

`np.concatenate((arr1,arr2),axis=0)`	Adds arr2 as rows to the end of arr1
`np.concatenate((arr1,arr2),axis=1)`	Adds arr2 as columns to end of arr1
`np.split(arr,3)`	Splits arr into 3 sub-arrays
`np.hsplit(arr,5)`	Splits arr horizontally on the 5th index

Indexing/slicing/subsetting

`arr[5]`	Returns the element at index 5
`arr[2,5]`	Returns the 2D array element on index [2][5]
`arr[1]=4`	Assigns array element on index 1 the value 4
`arr[1,3]=10`	Assigns array element on index [1][3] the value 10
`arr[0:3]`	Returns the elements at indices 0,1,2 (On a 2D array: returns rows 0,1,2)
`arr[0:3,4]`	Returns the elements on rows 0,1,2 at column 4
`arr[:2]`	Returns the elements at indices 0,1 (On a 2D array: returns rows 0,1)
`arr[:,1]`	Returns the elements at index 1 on all rows
`arr<5`	Returns an array with boolean values
`(arr1<3) & (arr2>5)`	Returns an array with boolean values
`~arr`	Inverts a boolean array
`arr[arr<5]`	Returns array elements smaller than 5

Vector Math

`np.add(arr1,arr2)`	Elementwise add arr2 to arr1
`np.subtract(arr1,arr2)`	Elementwise subtract arr2 from arr1
`np.multiply(arr1,arr2)`	Elementwise multiply arr1 by arr2
`np.divide(arr1,arr2)`	Elementwise divide arr1 by arr2
`np.power(arr1,arr2)`	Elementwise raise arr1 raised to the power of arr2
`np.array_equal(arr1,arr2)`	Returns True if the arrays have the same elements and shape
`np.sqrt(arr)`	Square root of each element in the array
`np.sin(arr)`	Sine of each element in the array
`np.log(arr)`	Natural log of each element in the array
`np.abs(arr)`	Absolute value of each element in the array
`np.ceil(arr)`	Rounds up to the nearest int
`np.floor(arr)`	Rounds down to the nearest int
`np.round(arr)`	Rounds to the nearest int

Scalar Math

`np.add(arr,1)`	Add 1 to each array element
`np.subtract(arr,2)`	Subtract 2 from each array element
`np.multiply(arr,3)`	Multiply each array element by 3
`np.divide(arr,4)`	Divide each array element by 4 (returns np.nan for division by zero)
`np.power(arr,5)`	Raise each array element to the 5th power

Statistics

`np.mean(arr,axis=0)`	Returns mean along specific axis
`arr.sum()`	Returns sum of arr
`arr.min()`	Returns minimum value of arr
`arr.max(axis=0)`	Returns maximum value of specific axis
`np.var(arr)`	Returns the variance of array
`np.std(arr,axis=1)`	Returns the standard deviation of specific axis
`arr.corrcoef()`	Returns correlation coefficient of array

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